

SRAI

# SETS, LOGIC, RELATIONS AND FUNCTIONS

Book 1 · Lesson 2 · PU-B01-C02

*Audited controlled manuscript aligned with M1\_N02\_sets\_logic\_relations\_functions.ipynb*

*A transparent intelligent system begins with an explicit universe, explicit relations, and testable logical rules.*

| Control field      | Value                                                                         |
|--------------------|-------------------------------------------------------------------------------|
| Owner and author   | Mbaye Kebe                                                                    |
| Canonical sequence | Book 1, Chapter 2; follows Mathematical Thinking and Reproducible Computation |
| Companion notebook | M1_N02_sets_logic_relations_functions.ipynb                                   |
| Status             | Audited release candidate; Colab validation required before publication       |

# 1. Defining the problem before automating it

Before a system can reason correctly, it must know which objects belong to the problem, which statements are true, which entities are connected, and how an input maps to an output. Sets, logic, relations and functions provide this formal language.

## 1.1 Learning objectives

1. Perform and interpret finite-set operations relative to an explicitly declared universe.
2. Construct truth tables and use De Morgan's laws to verify logical equivalence.
3. Distinguish implication from causation.
4. Represent relations as ordered pairs, matrices and graphs.
5. Distinguish arbitrary relations, partial functions and total functions.
6. Translate policy language into versioned, testable predicates.
7. Explain how these concepts support databases, knowledge graphs and accountable AI systems.

## 1.2 Opening case — an eligibility rule

A government introduces an emergency-assistance programme. A household qualifies if it is poor or food-insecure, has not already received equivalent assistance, and is located in an eligible district.

A computer cannot implement poor, food-insecure, equivalent assistance or eligible district without operational definitions. Let  $U$  be the universe of considered households,  $P$  the poverty set,  $F$  the food-insecurity set,  $B$  the prior-beneficiary set and  $G$  the eligible-geography set.

$$E = (P \cup F) \cap B^c \cap G, \quad \text{where } B^c = U \setminus B$$

The formula exposes the policy's structure. It also exposes its dependencies: the universe, the definition and update date of every set, the treatment of missing values, and the precedence of conflicting records.

# 2. Sets and the universe

A set is a collection of distinct objects. Membership is written  $x \in A$ . The singleton  $\{x\}$  is itself a set; therefore  $x \in A$  and  $\{x\} \subseteq A$  refer to different objects even though they have the same truth value.

$$A \cup B = \{x : x \in A \text{ or } x \in B\}$$

$$A \cap B = \{x : x \in A \text{ and } x \in B\}$$

$$A \setminus B = \{x : x \in A \text{ and } x \notin B\}$$

$$A^c = U \setminus A$$

$$A \times B = \{(a,b) : a \in A \text{ and } b \in B\}$$

A complement is meaningless without its universe. ‘Not vaccinated’ may refer to registered adults, all residents or everyone physically present. Different universes produce different complements and potentially different decisions.

### 3. Logic and truth conditions

For propositions P and Q,  $\neg P$  means NOT P,  $P \wedge Q$  means P AND Q, and  $P \vee Q$  means P OR Q. Material implication is defined by  $P \rightarrow Q \equiv \neg P \vee Q$ . It is false only when P is true and Q is false.

| P | Q | $P \wedge Q$ | $P \vee Q$ | $P \rightarrow Q$ |
|---|---|--------------|------------|-------------------|
| T | T | T            | T          | T                 |
| T | F | F            | T          | F                 |
| F | T | F            | T          | T                 |
| F | F | F            | F          | T                 |

$$\neg(P \wedge Q) \equiv (\neg P) \vee (\neg Q)$$

$$\neg(P \vee Q) \equiv (\neg P) \wedge (\neg Q)$$

Logical implication does not establish causation. It supplies neither temporal order nor a mechanism, intervention, counterfactual comparison or control of confounding.

### 4. Relations and functions

A binary relation from A to B is any subset  $R \subseteq A \times B$ . A patient may be related to a clinic, a farm to a district, an enterprise to a sector, or a document to a topic.

| Object                             | Formal condition                              | Interpretation                         |
|------------------------------------|-----------------------------------------------|----------------------------------------|
| Relation                           | $R \subseteq A \times B$                      | Any declared set of input–output pairs |
| Partial function                   | At most one output for each represented input | Some elements of A may be unmapped     |
| Total function $f:A \rightarrow B$ | Exactly one output in B for every $a \in A$   | Every declared input is mapped         |
| Injective function                 | Distinct inputs have distinct outputs         | No output is shared                    |
| Surjective function                | Every $b \in B$ is attained                   | Range equals codomain                  |

The relation  $\{(a,1),(b,2)\}$  is a total function only when the declared domain is  $\{a,b\}$ . The relation  $\{(a,1),(a,2)\}$  is not a function because a has two outputs. Declaring the domain and codomain is therefore part of the mathematical object, not optional metadata.

## 4.1 Matrix and graph representations

For  $A=\{a_1,\dots,a_m\}$  and  $B=\{b_1,\dots,b_n\}$ , a finite relation can be represented by a binary matrix  $M$  where  $M_{ij}=1$  when  $(a_i,b_j)\in R$  and 0 otherwise. The same structure may be viewed as a bipartite graph. Dense matrices require  $O(|A| \cdot |B|)$  memory; sparse edge lists or adjacency structures are preferable for large sparse relations.

## 5. Predicates, databases and knowledge systems

A predicate is a Boolean-valued function. In a database, a predicate becomes a filter: rows that do not satisfy a WHERE condition are eliminated from the result. Relations support joins, while ontologies represent individuals, classes and properties connecting entities.

| Mathematics       | Operational interpretation              |
|-------------------|-----------------------------------------|
| Set               | Table or result set                     |
| Membership        | A record satisfies a declared condition |
| Intersection      | Records satisfying both conditions      |
| Difference        | Exclusion                               |
| Cartesian product | All possible row pairs before filtering |
| Relation          | Link or join between entities           |
| Predicate         | Boolean filter or WHERE condition       |

Natural-language labels such as vulnerable or high risk must not be translated directly into code. They require an approved operational definition, provenance, effective date, missing-data rule and review process.

## 6. Fully worked example — vaccination prioritisation

Suppose priority is defined as age at least 65, healthcare-worker status, or clinical vulnerability, excluding anyone already fully vaccinated. Let  $A$ ,  $H$ ,  $V$  and  $F$  denote the respective sets within a declared registry universe  $U$ .

$$E = (A \cup H \cup V) \cap F^c$$

| Citizen | Age $\geq 65$ | Health worker | Vulnerable | Fully vaccinated | Eligible |
|---------|---------------|---------------|------------|------------------|----------|
| Aminata | Yes           | No            | No         | No               | Yes      |
| Cheikh  | No            | Yes           | No         | No               | Yes      |
| Mariama | No            | No            | Yes        | Yes              | No       |
| Ousmane | No            | No            | No         | No               | No       |

1. Declare  $U$  and confirm that every evaluated citizen belongs to  $U$ .
2. Evaluate the three inclusion predicates and take their union.
3. Evaluate the fully-vaccinated exclusion set and compute its complement relative to  $U$ .

4. Intersect inclusion with the complement of the exclusion set.
  5. Record the rule version and generate a reason for every acceptance or rejection.
- The resulting eligible set is {Aminata, Cheikh}. Mariama satisfies an inclusion condition but is removed by the exclusion rule. Ousmane satisfies no inclusion condition. These explanations are as important as the final Boolean output.

## 7. AI reasoning clinic

Claim: ‘If P implies Q, then P causes Q.’

Evaluation: False.  $P \rightarrow Q$  is a statement inside a formal logical system. A causal conclusion requires additional assumptions and evidence concerning interventions, temporal order, mechanisms and alternative explanations.

Improved reasoning: ‘Within the declared rule system, every case satisfying P must satisfy Q. Whether changing P would change Q is a separate causal question.’

## 8. Failure modes and safeguards

| Failure mode                             | Required safeguard                                       |
|------------------------------------------|----------------------------------------------------------|
| Undefined universe                       | Declare U and test all evaluated records for membership  |
| Ambiguous category                       | Approve an operational definition and record its version |
| Partial mapping treated as total         | Test coverage of the entire declared domain              |
| Implication treated as causation         | Separate logical validity from causal evidence           |
| Missing values silently treated as false | Specify three-valued or explicit missing-data handling   |
| Dense relation structure at scale        | Use sparse representations and document complexity       |
| Rule changed without traceability        | Version code, data, rules and effective dates            |

## 9. Laboratory and exercises

1. Verify both De Morgan laws with complete truth tables.
2. Explain the difference between  $x \in A$  and  $\{x\} \subseteq A$ .
3. For  $A=\{1,2,3\}$  and  $B=\{a,b\}$ , enumerate  $A \times B$ .
4. Convert a relation matrix into ordered pairs and back again.
5. Design tests that distinguish an arbitrary relation, a partial function and a total function.
6. Build a programme-eligibility engine with inclusion and exclusion sets, explicit missing-data treatment and reasons for every decision.

7. Run M1\_N02 from a fresh Google Colab runtime using only the controlled PU-B01-C02 release ZIP; reproduce every assertion and save the environment manifest.

## 10. Student assessment and instructor solutions

| Assessment question                                         | Controlled answer                                                                         |
|-------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| State both De Morgan laws for sets.                         | $(A \cup B)^c = A^c \cap B^c$ and $(A \cap B)^c = A^c \cup B^c$ , relative to $U$ .       |
| When does $R \subseteq A \times B$ define a total function? | Every $a \in A$ is paired with exactly one $b \in B$ .                                    |
| Give an invalid implication pattern.                        | Affirming the consequent: $P \rightarrow Q$ and $Q$ do not entail $P$ .                   |
| Distinguish codomain from range.                            | The codomain is the declared destination set; the range is the subset actually attained.  |
| Why do types and domains matter in AI pipelines?            | They make admissible inputs, outputs and operations explicit and prevent category errors. |

Scoring guide: 4 = correct result with valid reasoning; 3 = substantially correct reasoning; 2 = partial structure; 1 = relevant attempt; 0 = no relevant evidence.

## 11. Key insight and SRAI principle

Sets determine what belongs. Logic determines truth conditions. Relations determine what is connected. Functions determine how declared inputs map to outputs. Together they form the formal language of data filters, databases, knowledge graphs, algorithms and transparent decision systems.

*SRAI Principle No. 2: A transparent intelligent system begins with an explicit universe, explicit relations, and testable logical rules.*

## 12. Looking ahead

Chapter 3 turns from membership and logic to geometry. Vectors represent observations, model parameters, gradients and decision states as ordered quantities that can be compared, projected and transformed.

## 13. Verified references

- OpenStax (2023). Contemporary Mathematics, §§1.1, 1.4–1.5: basic set concepts and set operations. <https://openstax.org/books/contemporary-mathematics/pages/1-introduction>
- Python Software Foundation. Python 3 documentation, §5.4 Sets: union, intersection, difference and symmetric difference. <https://docs.python.org/3/tutorial/datastructures.html#sets>
- PostgreSQL Global Development Group. PostgreSQL SELECT and table-expression documentation: Boolean WHERE conditions and row filtering. <https://www.postgresql.org/docs/current/sql-select.html>

- W3C (2012). OWL 2 Web Ontology Language Primer, Second Edition: individuals, classes and relations as properties. <https://www.w3.org/TR/owl2-primer/>
- National Academies of Sciences, Engineering, and Medicine (2019). Reproducibility and Replicability in Science. <https://doi.org/10.17226/25303>

## 14. Release gate

- Owner substantive review completed.
- Mathematical definitions checked against the declared domain and codomain.
- M1\_N02 executed locally and from a clean Google Colab runtime.
- Complete solutions and notebook outputs included.
- Website, YouTube, GitHub and LinkedIn assets derived only from this controlled manuscript and validated notebook.
- Version inventory, checksums and release record generated before publication.